

EX: 6 Create and Android application to implement a Start-Stop counter using buttons.

Activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>

<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:gravity="center"
    android:orientation="vertical"
    android:padding="20dp">

    <TextView
        android:id="@+id/tvCounter"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="0"
        android:textSize="40sp"
        android:textStyle="bold"
        android:layout_marginBottom="30dp"/>

    <Button
        android:id="@+id/btnStart"
        android:layout_width="200dp"
        android:layout_height="wrap_content"
        android:text="Start"/>

    <Button
        android:id="@+id/btnStop"
        android:layout_width="200dp"
```

```
        android:layout_height="wrap_content"
        android:text="Stop"
        android:layout_marginTop="10dp"/>
<Button
    android:id="@+id/btnReset"
    android:layout_width="200dp"
    android:layout_height="wrap_content"
    android:text="Reset"
    android:layout_marginTop="10dp"/>
</LinearLayout>
```

MainActivity.kt

```
package com.example.counterapp

import android.os.Bundle
import android.os.Handler
import android.os.Looper
import androidx.appcompat.app.AppCompatActivity
import android.widget.Button
import android.widget.TextView

class MainActivity : AppCompatActivity() {
    private lateinit var tvCounter: TextView
    private lateinit var btnStart: Button
    private lateinit var btnStop: Button
    private lateinit var btnReset: Button
    private var counter = 0
    private var isRunning = false
    private val handler = Handler(Looper.getMainLooper())
```

```

private val runnable = object : Runnable {
    override fun run() {
        if (isRunning) {
            counter++
            tvCounter.text = counter.toString()
            handler.postDelayed(this, 1000)
        }
    }
}

override fun onCreate(savedInstanceState: Bundle?) {
    super.onCreate(savedInstanceState)
    setContentView(R.layout.activity_main)
    tvCounter = findViewById(R.id.tvCounter)
    btnStart = findViewById(R.id.btnStart)
    btnStop = findViewById(R.id.btnStop)
    btnReset = findViewById(R.id.btnReset)
    btnStart.setOnClickListener {
        if (!isRunning) {
            isRunning = true
            handler.post(runnable)
        }
    }
    btnStop.setOnClickListener {
        isRunning = false
    }
    btnReset.setOnClickListener

```

```
        isRunning = false
        counter = 0
        tvCounter.text = "0"
    }
}
}
```

OUTPUT

